

英文 1 學習卷

Lesson 4 Sniffing Out More Than Just Bones

____年____班____號 姓名：_____

總分

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限定用途教學教材，不得公開販售或流通

I. 文意字彙與詞類變化：30%

- predict 1. Some people claim that they can p_____t what will happen tomorrow, but I think it's impossible to know the future.
- rewards 2. Joe trains his dog by giving him treats as r_____ds for being good.
- behavior 3. Cesar Millan, a famous dog trainer, once said that owners should understand the meaning of their dogs' b_____r because what dogs do explains their needs and moods.
- exposure 4. Make sure you're protected from the sun; even brief e_____e to the midday sun can cause sunburn.
- item 5. The teacher asked the students to bring an i_____m that represented their culture for the school presentation.
- complete 6. Your report is not c_____e. You forgot the introduction.
- performing 7. The magician is rehearsing an amazing trick that he will be p_____ming for the audience at the show tonight.
- creative 8. To be a successful novelist, you need to be very _____ (create) and come up with a lot of new ideas.
- infected 9. The patient was isolated because the doctor didn't want other people to be _____ (infect).
- assignments 10. The teacher gave her students a lot of reading _____ (assign) to do during winter vacation.

II. 文法選擇：10%

- (B) 11. I was scared while hiking up the mountain yesterday because we were only _____ feet away from the edge of the cliff.
(A) a little (B) a few (C) many (D) much
- (C) 12. The ice in the Arctic keeps melting, and _____ this reason, polar bears are forced to travel farther to find food.
(A) as (B) of (C) for (D) to
- (D) 13. My roommates had a bitter quarrel last weekend, and the result is _____ they haven't talked to each other for days now.
(A) how (B) which (C) this (D) that
- (D) 14. Fiona mastered French cuisine _____ watching cooking videos on YouTube.
(A) on (B) upon (C) for (D) by
- (A) 15. The risk of smoking can be _____ greater than imagined if you have a weak heart.
(A) far (B) so (C) very (D) really

III. 綜合測驗：10%

I'm a sniffer dog trainer. We can train these dogs to do some amazing things. In Pingtung, sniffer dogs 16. trees with a disease called brown root rot. When a dog finds a sick tree, it sits next to the tree and waits for a trainer like me to give it a reward. Then, workers come and dig out the 17. tree before it falls and hurts someone. This method has been so successful that parks all over Taiwan are starting to use sniffer dogs too.

Because of their superior olfactory* sensitivity, 18. is not surprising that sniffer dogs can not only find sick trees but also help to predict when certain problems are about to affect humans. I remember one woman, Maria Tello 19. had diabetes. One night, her dog Malachi woke up. He could smell that her blood sugar was dangerously low and woke her up. He then ran to get her medicine. Malachi actually saved Maria's life five times that year!

So how do these dogs do it? Diseases create chemical changes in plants, animals, and even humans. These chemical changes create special smells, and we trainers familiarize* the dogs with these smells over and over again. Then, we teach the dogs a specific action to 20. when they detect these smells. After the dogs are trained, they're ready to work. Pretty cool, huh?

註：olfactory 嗅覺的 familiarize 使……對某物熟悉

- (A) 16. (A) are used to help find (B) need to be exposed to
(C) are assigned to carry out (D) manage to accomplish
- (A) 17. (A) infected (B) alert (C) keen (D) sensitive
- (D) 18. (A) that (B) this (C) what (D) it
- (B) 19. (A) who (B) , who (C) that (D) , which
- (C) 20. (A) let out (B) turn to (C) carry out (D) go through

IV. 文意選填：10%

(A) sensitive	(B) delicious	(C) manage	(D) drugs	(E) behave
(F) locate	(G) wonder	(H) tasks	(I) keener	(J) ability

Dogs are well known for their sense of smell. But did you know that pigs' noses may be even more 21. than dogs'? It has been proved that pigs have many more of the cells that are used for smelling. Pigs are able to 22. food up to eleven kilometers away or even seven meters underground! French farmers take advantage of this 23. to find valuable truffles* that grow wild in the forest. Pigs hunt for the truffles, and once they find them, they quickly dig up the 24. treat.

You might 25., since pigs are such great sniffers, why they aren't used by the police or at airports. Actually, some police departments have used pigs to detect 26.. However, pigs are much more difficult to 27. than dogs, so this practice is not common. Pigs may have a(n) 28. sense of smell than dogs, but they are not as easy to train. A pig might not 29. well in a crowd of people. That's why such 30. are best left to dogs. Pigs are probably much happier in the forest anyway.

註：truffle 松露

21. A 22. F 23. J 24. B 25. G
26. D 27. C 28. I 29. E 30. H

V. 閱讀測驗：9%

Did you know that dogs can smell things 10,000 times better than humans? That may seem impossible, but it’s true. This great sense of smell makes dogs great partners for policemen. Dogs can be trained to sniff out all kinds of things, like drugs, bombs, cash, and food.

They can even smell what is hidden inside other things because they can smell in layers*. For example, they can sniff out a rubber ball wrapped with a smelly basketball shirt and closed inside a wooden box. Isn’t that incredible?

Detection dogs are not new. In fact, dogs have been used to keep people safe for thousands of years and through several wars. These dogs not only have a great sense of smell. They can also recognize voices and even the shape of a person’s shadow.

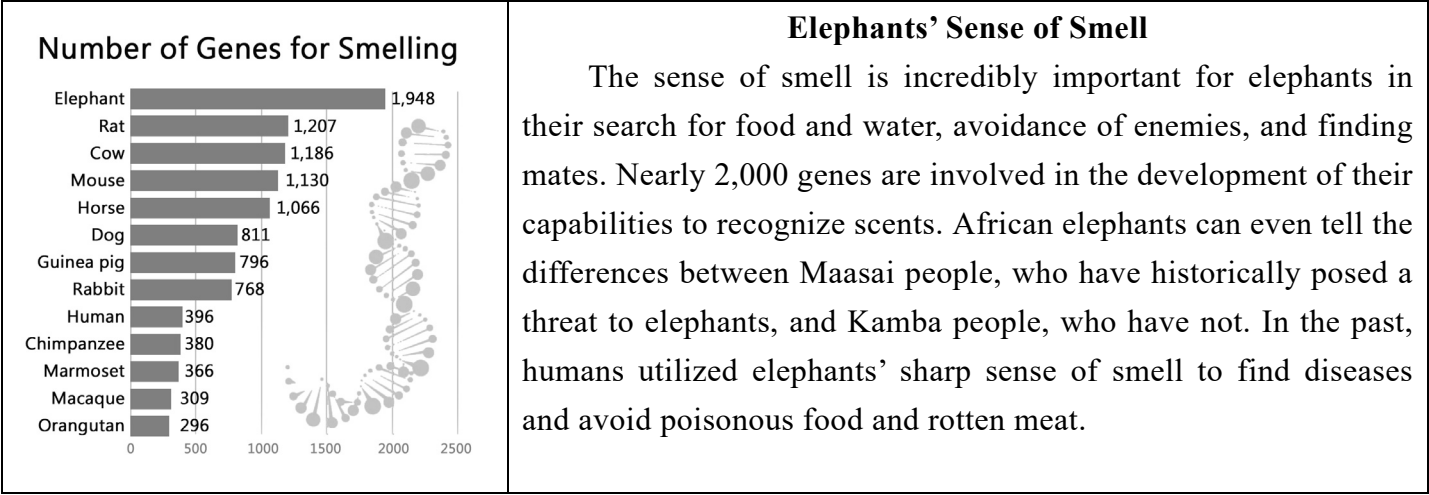
Detection dogs are used in airports to sniff out dangerous substances* and fruits and vegetables that could be bad for local ecosystems*. Before you get on a plane, make sure you don’t bring any fruit with you, or you may have an unwanted dog friend!

註：layer 分層 substance 物質 ecosystem 生態系統

- (B) 31. What is the purpose of this passage?
- (A) To advise people to avoid detection dogs.
- (B) To introduce detection dogs to people.
- (C) To ask people to protect detection dogs.
- (D) To record the training of detection dogs.
- (D) 32. Which of the following descriptions is **NOT** mentioned about detection dogs?
- (A) They can find people based on their smell and voice.
- (B) They can smell drugs in closed luggage or backpacks.
- (C) They are very useful to policemen and soldiers.
- (D) People started to use them just a few years ago.
- (C) 33. Why does the author advise people against bringing fruit onto a plane?
- (A) Fruit can be a dangerous bomb.
- (B) Detection dogs will get sick.
- (C) Fruit will attract detection dogs.
- (D) People need to adopt detection dogs.

VI. 混合題：11%

We know that dogs have a good sense of smell; however, according to a study conducted by Dr. Yoshihito Niimura at the University of Tokyo, many other animals have a better sense of smell than dogs. If you are interested in comparing the sense of smell of certain animals, the following chart may provide you with some useful information:



34. What is illustrated by the chart with respect to the animal kingdom? Choose **TWO** answers. (4%)
- (B) (F)
- (A) The chart ranks different animals based on their vision.
- (B) Elephants have the most genes for smelling, among the animals listed.
- (C) Horses have more genes related to smell than cows do.
- (D) Humans have more genes for smelling than guinea pigs.
- (E) Dogs are at the top of the list for the number of genes for smelling.
- (F) The number of genes for smelling in chimpanzees is slightly lower than in humans.
35. Which word in the passage means “made good use of”? (4%)
- utilized
36. 下列簡短敘述摘記上方文章重點。請從文章中找出最適當的單詞 (word) 填入下列句子空格中，使句子語意完整、語法正確，並符合全文文意。(3%)
- Elephants have a highly developed sense of smell that enables them to recognize different scents, including avoiding poisonous food and telling apart possible threats.

VII. 翻譯：20% (37 至 39 每格 1.5 分，40 至 41 每題 4 分)

37. 這個隊伍在兩分鐘之內便兩次射門得分，突然之間，他們看似勝券在握將會贏得這場比賽。
- The team scored two goals in two minutes, and all of a sudden, it looked as if they would win the match.
38. 全球氣溫上升普遍認為是因為大氣中溫室氣體含量增加所致。
- The rise in the global temperature is believed to be due to the increased levels of greenhouse gases in the atmosphere.
39. 這項實驗的結果與科學家們所假設的不同。
- The experiment turned out differently than the scientists had assumed.
40. 這就是我媽媽幫我弟弟買的那個鉛筆盒。(形容詞子句)
- This is the pencil case that/which my mom bought for my little brother.
41. Jimmy 的家鄉在以它的高粱酒 (Kaoliang liquor) 而聞名的金門 (Kinmen)。(形容詞子句)
- Jimmy’s hometown is in Kinmen, which is famous for its Kaoliang liquor.

英文 1 學習卷詳解——第 6 回

Lesson 4 Sniffing Out More Than Just Bones

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I. 文意字彙與詞類變化

1. predict	2. rewards	3. behavior	4. exposure	5. item
6. complete	7. performing	8. creative	9. infected	10. assignments

1. 有些人聲稱他們可以預測明天會發生什麼事，但我認為要預知未來是不可能的。
2. Joe 透過給予零食作為表現良好的獎勵來訓練他的狗。
3. 著名的馴犬師 Cesar Millan 曾說過，主人應該理解狗狗行為背後的意義，因為狗狗所做的事解釋了牠們的需求與情緒。
4. 確保你有做好防曬；即便只是短暫地暴露在正午的陽光下，也可能導致晒傷。
5. 老師要求學生帶一件代表自己文化的物品來進行學校報告。
6. 你的報告不完整。你忘了寫前言。
7. 魔術師正在排練一個驚人的戲法，他將在今晚的表演中為觀眾演出。
8. 要成為一名成功的小說家，你需要非常有創意並想出許多新點子。
9. 該名病患被隔離了，因為醫生不希望其他人被感染。
10. 老師給了學生許多閱讀作業，讓他們在寒假期間完成。

II. 文法選擇

11. (B)	12. (C)	13. (D)	14. (D)	15. (A)
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11. a little + 不可數名詞、a few + 可數名詞，兩者皆表示一些，但若前面再加上 only，則表示很少；many + 可數名詞、much + 不可數名詞，兩者皆表示很多。故依題意，選(B)，表示距離懸崖邊僅幾英尺遠。
12. for this reason 因為這個原因
13. that 引導名詞子句，作為主詞補語。
14. by V-ing 藉由……的方法
15. 常用以修飾比較級的副詞有(so) much、even、rather、far、a little、no 等，far 表示「遠遠；非常」。

III. 綜合測驗

16. (A)	17. (A)	18. (D)	19. (B)	20. (C)
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我是一名嗅探犬訓練師。我們可以訓練這些狗做一些了不起的事情。在屏東，嗅探犬 ¹⁶用來幫助尋找患有稱作「褐根病」的樹木。當狗兒發現一棵病樹時，牠會坐在該棵樹旁邊，並等待像我這樣的訓練師給牠獎勵。然後，工人們會在 ¹⁷受感染的樹倒下傷到人之前將其挖出。這種方法非常成功，臺灣各地的公園也都開始使用嗅探犬。

因為牠們超群的嗅覺靈敏性，嗅探犬不僅能找到生病的樹木，還可以幫助預測某些問題何時會影響到人類，¹⁸這點也就不令人意外了。我記得一位女士名叫 Maria Tello，¹⁹她患有糖尿病。一天晚上，她的狗 Malachi 醒來了。牠聞到她的血糖低到了會有危險的程度，就把她叫醒，然後跑去給她拿藥。Malachi 那一年實際上救了 Maria 五次！

那麼這些狗兒是怎麼做到的呢？疾病會在植物、動物甚至人類身上造成化學變化。這些化學變化會產生特殊的氣味，我們訓練員會一遍又一遍地讓狗兒熟悉這些氣味。然後，我們教牠們聞到這些氣味時要 ²⁰執行的特定動作。狗兒受訓過後，就準備好開始工作了。很酷吧？

16. 主詞是嗅探犬 (sniffer dogs)，牠們是被「使用／利用」來協助尋找樹木。be used to + VR 表「被用來做某事」。(B) 需要被暴露於；(C) 被指派去執行；(D) 設法完成。
17. 前文提到樹木生病了 (sick tree)，因此這裡指「被感染的」(infected) 樹木。(B) 警覺的；(C) 敏銳的；(D) 敏感的
18. 真正的子句主詞是 that sniffer dogs can...，但因為太長，所以句首用虛主詞 it 代替。句型為 It + be + adj. + that + 子句。
19. 先行詞為人名，故不需加以限定，而選「, who」。
20. 根據上下文，訓練師會教導狗狗在聞到味道時，「執行」(carry out) 一個特定的動作。(A) 發出（聲音）；(B) 轉向／求助；(D) 經歷。

IV. 文意選填

21. (A)	22. (F)	23. (J)	24. (B)	25. (G)	26. (D)	27. (C)	28. (I)	29. (E)	30. (H)
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狗兒以其嗅覺而聞名。但是你知道豬的鼻子可能甚至比狗的鼻子更 ²¹靈敏嗎？已證實豬有更多用於嗅探的細胞。豬能夠 ²²找到距離十一公里甚或地下七公尺處的食物！法國農民善用了這種 ²³能力來尋找野生於森林中有價值的松露。豬會尋找松露，牠們一旦找到，就會迅速挖出這些 ²⁴美味的食物。

你可能 ²⁵會想知道，既然豬是如此出色的嗅探者，為什麼警方或機場不善用牠們呢？事實上，一些警察部門已經用豬來偵查 ²⁶毒品。不過，豬比狗難 ²⁷管理多了，所以這種作法並不常見。豬可能比狗有 ²⁸更敏銳的嗅覺，但牠們並不容易訓練。豬在人群中可能無法 ²⁹表現得很守規矩。這就是為什麼這樣的 ³⁰任務最好還是留給狗兒。無論如何，豬在森林裡可能會更快樂。

take advantage of 善用……

21. 此處需填形容詞修飾鼻子，前句提到狗的嗅覺很有名，後句用 even more 加強語氣，指豬的鼻子可能更「敏感」。故選 (A) sensitive。
22. 此處需填原形動詞，接在 be able to 之後，語意為豬能透過嗅覺「找到」極遠處或地底下的食物。故選(F) locate。
23. 此處需填名詞，take advantage of this...句意為「善用這種（嗅覺）能力」來尋找昂貴松露。故選(J) ability。
24. 此處需填形容詞修飾 treat，指松露對豬而言是「美味的」食物。故選 (B) delicious。
25. 此處需填原形動詞，You might...句意為「你可能會納悶／想知道」，藉此引出後方關於警察不常使用豬的疑問。故選(G) wonder。
26. 此處需填名詞作為 detect 的受詞，句意為警察部門曾利用豬來「偵測毒品」。故選(D) drugs。
27. 此處需填原形動詞，接在 be difficult to 之後，說明豬比狗更難以「管理」。故選(C) manage。
28. 此處需填形容詞比較級，後方有 than 做比較，語意為豬的嗅覺比起狗「更敏銳」。故選(I) keener。
29. 此處需填動詞，behave well 為固定搭配，指豬在人群中的「表現」或「行為」可能不佳。故選(E) behave。
30. 此處需填複數名詞作為主詞，指代前文提到的搜索工作，句意為「這類任務最好還是留給狗去做」。故選(H) tasks。

V. 閱讀測驗

31. (B)	32. (D)	33. (C)
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你知道狗的嗅覺比人類好一萬倍嗎？這或許聽起來像是不可能，但卻是事實。這個了不起的嗅覺讓狗成為警方的偵查好幫手。狗可以被訓練來聞出各種東西像是毒品、炸彈、現金跟食物。

狗甚至可以聞到藏在其他東西裡面的物品，因為牠們可以聞到被層層包覆的東西。例如說，牠們可以聞到包在發臭的籃球衣中並塞在木頭盒子裡的橡皮球。這是不是很令人難以置信？

偵查犬不是新的概念。事實上，幾千年來以及在幾次戰爭中，狗一直被用來維持人類安全。這些狗不只擁有敏銳的嗅覺，還能辨識聲音和甚至一個人的影子。

偵查犬在機場被用來聞出危險物質和可能會對本土生態有害的水果蔬菜。當你搭飛機之前，務必確認你沒有帶水果，不然你可能會有不想要的狗朋友！

rubber ball 橡皮球；wrap 包起來；incredible 難以置信的

31. 此處考驗全文主旨。文章從狗的超強嗅覺出發，描述其協助警察偵查、嗅出層層包裹的物品，以及在機場維持安全的歷史與功能，旨在向讀者「介紹」偵查犬。故選(B)。

32. 此處需選出「未被提及」的描述。由第三段第一句「Detection dogs are not new.」以及隨後提到「已使用數千年」可知，偵查犬並非這幾年才出現，故選(D)。

33. 最後一段提到，偵查犬會嗅出可能破壞生態系統的蔬菜水果，並俏皮地警告如果不小心，可能會引來一位「不想要的狗狗朋友」。意即水果會「吸引」偵查犬過來查驗。故選(C)。

VI. 混合題

34. (B) (F)	35. utilized	36. recognize
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我們都知道狗擁有良好的嗅覺；然而，根據東京大學新村芳人（Yoshihito Niimura）博士所進行的一項研究顯示，許多其他動物的嗅覺其實比狗的還要好。如果你對比較特定動物的嗅覺感興趣，下方的圖表或許能為你提供一些有用的資訊：

嗅覺基因數量	
大象（Elephant）：1,948	對大象而言，嗅覺在尋找食物與水源、躲避敵人以及尋找配偶方面極其重要。有將近 2,000 個基因參與了牠們辨識氣味能力的發展。非洲象甚至能區分馬賽人（Maasai）與坎巴人（Kamba），前者在歷史上一直對大象構成威脅，而後者則沒有。在過去，人類曾利用大象敏銳的嗅覺來發現疾病並避開有毒食物以及腐爛的肉類。
老鼠（Rat）：1,207	
牛（Cow）：1,186	
小鼠（Mouse）：1,130	
馬（Horse）：1,066	
狗（Dog）：811	
天竺鼠（Guinea pig）：796	
兔子（Rabbit）：768	
人類（Human）：396	
黑猩猩（Chimpanzee）：380	
狨猴（Marmoset）：366	
獼猴（Macaque）：309	
紅毛猩猩（Orangutan）：296	

綜合解析：此題組為雙文本呈現方式，需閱讀兩項資料並比較其中關聯始可作答。

34. (B)此處需觀察長條圖最高處，大象（Elephant）以 1,948 個基因位居清單之首，故正確；(F)此處需比較數據，黑猩猩（Chimpanzee, 380）確實比人類（Human, 396）稍微低一些，故正確。(A)錯誤：圖表是根據「嗅覺」而非「視覺」（vision）排名；(C)錯誤：馬（1,066）的基因數少於牛（1,186）；(D)錯誤：人類（396）遠少於天竺鼠（796）；(E)錯誤：狗（811）在圖表中僅排在中段，並非頂端。

35. 由定義可知應填入「動詞（且為過去式或過去分詞）」，由文本最後的敘述可知，人類 utilized 大象的感官來執行如疾病檢測等複雜任務，故「utilized」即是「善用」之義。

36. 此處需填入一個原形動詞，接在 enables them to 之後。根據文章第二段提到的「capabilities to recognize scents」（辨識氣味的能力），此處填入 recognize 最符合語意，表示大象高度發達的嗅覺能使牠們辨別不同的氣味。

VII. 翻譯

37. all；of；a；sudden
38. due；to
39. turned；out
40. This is the pencil case that/which my mom bought for my little brother.
41. Jimmy’s hometown is in Kinmen, which is famous for its Kaoliang liquor.

37. all of a sudden 突然間

38. due to 因為……

39. turn out 結果是

40. 此處使用「限定」形容詞子句後位修飾聽讀者未知的 the pencil case，「限定」形容詞子句可使用 that 或 which 修飾「非人」先行詞。

41. 因先行詞「Kinmen」（金門）獨一無二，聽讀者不會混淆，故使用毋須限定何者，「非限定」形容詞子句前方需加上逗號表示此為補充資訊，且僅能使用 which 修飾「非人」先行詞。